

MEDICINE
PRACTICALS
(OSCE,
X-RAYS,INSTRUMENTS)



OSLER'S ACADEMY

Prepared by Dr. DILIP KUMAR R MD, MRCP, for **Osler's Academy** Madhapur,
Hyderabad 81.

OSCE SET - 1 (15 nos) UNDERGRADUATES (with preference to Curriculum of
NTR University)
Difficulty - Easy

OSCE 1 :

A 21 year old lady is coming to you with complaints of lethargy, constipation and irregular menses . She also admits she has been intolerant to cold off late and that she gained a substantial amount of weight in recent times. On examination she was obese (BMI -31) , has a BP- 130/110 mm Hg, PR- 48/min , and a smooth butterfly shaped swelling in the anterior aspect of neck. Based on the data :

1. What is your first diagnosis ?
2. What blood test you will do to diagnose the above condition ?
3. Name two skin changes associated with the condition ?
4. Name two complications of the disease.
5. What is the treatment of choice for the above condition and how will you give ?

ANSWER

1. Hypothyroidism
2. Free T3, T4 and TSH
3. Alopecia , Dry skin , Periorbital puffiness
4. Myxedema coma, Bradyarrhythmia , Pericardial Effusion/ tamponade , Hypertension
5. Levothyroxine – once a day in the morning in empty stomach

OSCE 2 :

A 21 year old lady admitted with complaints of excessive tiredness, palpitations and sweating . She also admits that she has lost some weight in recent times and also agrees that she is intolerant to heat off late . On direct questioning she also admits that she had frequent episodes of diarrhea and irregular menses. On examination she had a PR- 121/min and regular , BP- 160/90 mm Hg and a smooth butterfly shaped swelling in the anterior aspect of neck. She was also found to have some degree of exophthalmos as well . Based on the data given :

1. What is your likely diagnosis ?
2. What is the most common cause of exophthalmos worldwide ?
3. Name two cardiac complications of the disease .
4. What will be the initial test to confirm your diagnosis ?
5. Name four drugs that can be used for treating the condition .

ANSWER

1. Hyperthyroidism
2. Grave's disease
3. Tachyarrhythmias (SVT), Cardiomyopathy
4. Free T3, T4 and TSH



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3. Systemic embolism (Stroke, mesenteric ischemia), Cardiogenic shock/Acute pulmonary edema .
4. DC cardioversion , Drugs (beta blockers , Digoxin , Calcium channel blockers, Amiodarone, anticoagulation – warfarin), Surgery- ex. AV node ablation
5. Thyrotoxicosis , Alcohol, COPD, Pulmonary embolism, old age.

OSCE 10:

A 30 year old gentleman comes to OPD with complaints of retrosternal discomfort of 4 days duration. He admits that he had retrosternal pain which was non radiating and associated with burning sensation especially in the night immediately after food. He had a fracture 1 week back and he was taking some pain killers prescribed by an orthopaedician . On examination he is well , vitals are stable and he had mild tenderness in the epigastric region.

1. What is the likely diagnosis and the cause in this patient ?
2. How will you treat and name any two drugs that can be used in treating this patient .
3. Name two potential complications of the condition.
4. Name two drugs that are active against H.pylori .
5. Name one investigation useful in confirming the diagnosis .

ANSWERS :

1. NSAID induced gastritis.
2. Stop NSAID , PPI(omeprazole, pantaprazole), H2 blockers(ranitidine) , Sucralfate .
3. Anaemia , UGI bleed(hematemesis , malena), gastric perforation.
4. Amoxixillin, Clarithromycin, metronidazole.
5. UGI endoscopy.

You are viewing this patient who has a five day history of fever and his Blood smear shows RBC's positive for *P. falciparum*.

1. Name the type of fever *P. falciparum* causes
2. Name two complications of the disease
3. Name two drugs that will be effective in treating *P. falciparum*
4. Name the drug that should be given very carefully in patients with G6PD deficiency.
5. Name two drugs that can be used for chemoprophylaxis of malaria.

1. Malignant tertian malaria
2. Black water fever , Cerebral malaria
3. Artemether , Artether ,
Lumefantrine+artemether , Quinine
4. Primaquine
5. Mefloquine , Doxycycline

A 26 year old lady comes to you with complaints of jaundice and fatigue of 3 months duration. On examination, patient was pale and had moderate splenomegaly. Investigations revealed, Hemoglobin- 3.1 g/dl, Serum Bilirubin – 4.8 mg/dl with 98 % being Indirect bilirubin . Reticulocyte count – 11.4% (normal – 0.5-1.5) .

1. What is your likely diagnosis?
2. Name two causes
3. Name two relevant investigations
4. Mention two causes of massive splenomegaly.
5. Name atleast two things that you would like to look in the peripheral smear of this patient.

1. Hemolytic anaemia
2. Thalessemia , Hereditary spherocytosis , Autoimmune Hemolytic anaemia , G6PD deficiency
3. Peripheral Smear , LDH, Hb electrophoresis
4. Kala azar , Malaria, Myelofibrosis , Hairy cell leukemia
5. Target cells , Spherocytes, Elliptocytes , Sick cells , nucleated RBC's.

STEPS :

1. Introduce yourself
2. Explain what you are going to do
3. Examine orbicularis frontalis and look for bell's phenomenon .
4. Examining deviation of angle of mouth .
5. Examining the integrity of buccinators (whistling , holding a pocket of air in the mouth against resistance)
6. Examining the platysma .
7. Thanking the patient .



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OSCE 15:

Examine this patient for signs for meningitis

STEPS :

1. Introduce yourself and explain what you are going to do
2. Eliciting neck stiffness (correct method using two hands plus rolling the neck side to side) and Brudzinski's
3. Eliciting photophobia and observe for pupils
4. Eliciting Kernig's sign
5. Thanking the patient .



ANSWERS :

1. Urinary Tract Infection (UTI)
2. E.coli
3. Urine C/S, Urine routine (for RBC's, pus cells, leukocyte esterase , nitrites)
4. Old age (post menopausal women), Diabetes mellitus , HIV , structural abnormalities of Urinary tract (ex. CKD, reflux disease)
5. Nitrofurantoin, Cotrimoxazole, Cipro/nor/levofloxacin, ceftriaxone, gentamicin.

OSCE 5 :

Examine this patient who has jaundice.

STEPS :

1. Introduce yourself and explain during each steps.
2. Look for eyes (sclera and conjunctiva- for icterus and pallor respectively)
3. See oral cavity (soft palate, under surface of tongue)
4. Signs of liver cell failure (parotids- enlargement/ hands- dupuytren's contracture and jaundice/ forearm ,elbows – scratch marks, palmar erythema / chest – spider naevi and gynaecomastia / legs – pedal edema)
5. Look for hepatomegaly , gall bladder and splenomegaly.
6. Look for left supraclavicular nodes .
7. Thank the patient for his/her cooperation.

OSCE 6 :

Examine this patient for signs of liver cell failure.

STEPS :

1. Introduce yourself and explain during each step.
2. Look for pallor and jaundice (eyes, oral cavity)
3. Face- parotids , temporal wasting.
4. Chest – gynaecomastia/breast atrophy, spider naevi
5. Hands- Clubbing , palmar erythema , Dupuytren's contracture.
6. Abdomen- Dilated veins , hepatosplenomegaly
7. Testicles – for atrophy
8. Legs- pedal edema .
9. Thank the patient.

OSCE 7 :

Examine this patient with anaemia .

STEPS :

1. Introduce yourself and explain at each step.
2. Look for pallor (lower palpebral conjunctiva, tip of tongue , soft palate, palms- especially creases)
3. Look for pulse (tachycardia)
4. Auscultate for murmurs and base of lungs for crepitations (failure)



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OSCE SET - 2 (10 nos) for UNDERGRADUATES (with preference to
Curriculum of NTR University)
Difficulty - Easy .

OSCE 1 :

A 16 year old girl who is a known case of type 1 diabetes admitted to the emergency with unconsciousness and seizures since half an hour . Her class mate told that she felt dizzy while talking and she was sweating before she collapsed and threw seizures and in the process she injured her left shoulder and forehead and was biting her tongue vigorously followed by which she had passed urine and faeces. She was taking her medicines (insulin) regularly and her mother also complained that she skipped her breakfast that day morning. On examination , she was *not* seizing actively now and she was deeply unconscious with response only to painful stimuli. Her cardiovascular , respiratory and other system examinations appear normal and her vitals are stable.

1. What is the likely diagnosis ?
2. What is the next investigation you would like to do to confirm the diagnosis ?
3. What is first treatment you want to give even before the investigation results (pertaining to the case) ?
4. Name two reasons from the history to tell that it is not pseudoseizure (hysteria)
5. Name two drugs used in the status epilepticus protocol.

ANSWERS :

1. Hypoglycemia .
2. Capillary Blood Glucose / Random Blood Sugar
3. IV 25% or 50% glucose.
4. External injuries, tongue bite, passing urine and stools.
5. Diazepam , Phenytoin, Phenobarbitone , Propofol, Midazolam .

OSCE 2 :

A 43 year old gentleman presents to you with fever, shortness of breath, cough with expectoration, right sided pricking chest pain which increases with inspiration for 2 days , all of which has worsened for the past 2 hours . He is a known case of type 2 diabetes of 4 years duration and his records suggest that he is not compliant with his diabetic medications with recent fasting blood sugar of 241 mg/dl. On examination he is febrile , and had increased vocal resonance, coarse crepitations with bronchial breath sounds in right infra axillary and infra scapular areas.

1. What is your diagnosis ?
2. Name two causative organisms.
3. What is the next best investigation.



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4. Name one complication which you should look for pertaining to this case.
5. If the patient is HIV positive with a CD4 count of 96/ μ l, what other organism you should cover for while starting treatment.

ANSWERS :

1. Right lobar pneumonia / Consolidation.
2. Pneumococcus, H.influenza, S.aureus, K.pneumoniae, Leigonella pneumophila.
3. Chest X ray PA view.
4. Diabetic ketoacidosis.
5. Pneumocystis jiroveci pneumonia . (PCP)

OSCE 3 :

A 56 year old gentleman comes to emergency with severe chest pain and breathlessness of 1 hour duration. The pain started when he was eating and is retrosternal and is radiating to his left shoulder, hand and jaw. On examination he looks unwell, sweating and he has a blood pressure of 78/30 mm Hg. He also had fine crepitations in both lung bases. You take an ECG which shows ST segment elevation (5mm) in precordial leads. (V1- V5) with reciprocal changes in inferior leads (II, III. aVF)

1. What is your diagnosis ?
2. What is the next investigation you would like to do to confirm the diagnosis ?
3. Name any two treatment modalities available.
4. Name two complications.
5. Name two drugs that will be beneficial in long term .

ANSWERS :

1. Acute coronary syndrome – STEMI
2. Troponin T/I or PCI (both are correct)
3. PCI, Thrombolysis
4. Cardiogenic shock, acute MR, VSD, Dressler's syndrome
5. Aspirin, statins (ex Atorvastatin), ACE inhibitors (ex.Enalapril), Beta blockers(ex. Metoprolol)

OSCE 4 :

A 45 year old lady comes to OPD with 2 day history of fever, burning sensation while passing urine and lower abdominal pain which radiates to the back. She is a known diabetic for >10 years and she on Metformin and glipizide for 6 years. Her recent Fasting blood sugar is 141 mg/dl and HbA1c is 7.1%. On examination, she is febrile and has mild tenderness on palpating the hypogastrium.

1. What is the likely diagnosis ?
2. What is the commonest organism ?
3. Name two investigations useful in this patient to confirm diagnosis .
4. Name two risk factors for the disease.
5. Name two drugs useful in treating the disease.

STEPS :

1. Introduce yourself
2. Explain what you are going to do
3. Examine orbito frontalis and look for bell's phenomenon .
4. Examining deviation of angle of mouth .
5. Examining the integrity of buccinators (whistling , holding a pocket of air in the mouth against resistance)
6. Examining the platysma .
7. Thanking the patient .

OSCE 13 :

Examine this patient looking for atleast 5 peripheral signs of Aortic regurgitation

STEPS :

1. Introduce yourself and explain about what you are going to do
2. Look for the following
 - Pupils
 - Oral cavity (uvula)
 - Carotid (corrigan's sign)
 - Nail bed (Quincke's)
 - Looking for water hammer pulse
 - Examining liver and spleen for pulsations
 - Auscultation of femorals (pistol shot, duroziez murmur)
3. Thanking the patient for cooperating

OSCE 14 :

A 26 year old lady comes to you with complaints of jaundice and fatigue of 3 months duration . On examination , patient was pale and had moderate splenomegaly. Investigations revealed , Hb- 3.1 g/dl , S.Bilirubin - 4.8 mg/dl with 98 % being Indirect bilirubin . Retic count - 11.4% (normal - 0.5-1.5) .

1. What is your likely diagnosis ?
2. Name two causes
3. Name two relevant investigations
4. Mention two causes of massive splenomegaly.
5. Name atleast two things that you would like to look in the peripheral smear of this patient .

ANSWER :

1. Hemolytic anaemia
2. Thalessemia , Hereditary spherocytosis , Autoimmune Hemolytic anaemia , G6PD deficiency
3. Peripheral Smear , LDH, Hb electrophoresis
4. Kala azar , Malaria, Myelofibrosis , Hairy cell leukemia
5. Target cells , Spherocytes, Elliptocytes , Sick cells , nucleated RBC's.



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5. Examine feet – for pedal edema . (failure)
6. Examine for jaundice and splenomegaly (hemolytic anaemia)
7. Thank the patient.

OSCE 8 :

A 48 year old gentleman comes to OPD for routine check up . He had no complaints except for occasional headaches. He is a non-smoker and non-alcoholic. His family history is non significant. He had no comorbidities till date. On examination he had no abnormalities except for a Blood pressure of 176/98 mm Hg. You diagnose him to be hypertensive.

1. What is the most common cause of hypertension in the world ?
2. Name two causes of secondary hypertension.
3. Name two complications of hypertension if left untreated.
4. Name two drug classes useful in treating the condition.
5. Name any two relevant / must tests you would like to do in this case.

ANSWERS :

1. Essential Hypertension.
2. Chronic kidney disease, Renal artery stenosis, Pheochromocytoma, Primary hyperaldosteronism(Conn's syndrome)
3. Hypertensive retinopathy, Chronic kidney disease, Cardiac failure /Pulmonary edema , Stroke , Peripheral vascular disease.
4. ACE inhibitors (enalapril), CCB's (amlodipine), Thiazides, Beta blockers.
5. Urine for protein, Fundus for retinopathy, ECG/Echo for LVH/Heart failure.

OSCE 9:

A 34 year old lady comes to you in emergency with complaints of shortness of breath and palpitations of 5 days duration. She is a known case of Mitral stenosis for 15 years and is on penicillin prophylaxis and she was previously well except for the last 5 days. On examination she is dyspneic, the pulse is feeble and irregular, BP- 100/60 mm Hg and auscultation revealed a mid diastolic murmur associated with a palpable diastolic thrill. She also has bi-basal crepitations in both lung bases.

1. What do you think is the likely cause of her recent decompensation ?
2. What are the characteristics of pulse and JVP you would expect in this patient ?
3. Name two important complications associated with the condition.
4. Name any two treatment modalities/drugs used to treat the condition.
5. Name two causes other than Valvular heart diseases .

ANSWERS :

1. Atrial fibrillation.
2. Irregularly irregular pulse, Varying volume, Pulse deficit >10 and no 'a' waves in JVP



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5. Carbimazole, methimazole , PTU, Propranolol , Steroids(Prednisolone) , Radioactive Iodine .

OSCE 3 :

Clinical examination of this patient with thyrotoxicosis with thyroid swelling

STEPS :

1. Introduce yourself
2. Examine nails (onycholysis)
3. Examine pulse (Tachycardia /AF)
4. Examine the eyes (exophthalmos)
5. Examine the extremities (for acropathy/ pretibial myxedema)
6. Inspect the gland from front (for enlargement)
7. Test with swallowing / drinking with water /protrusion of tongue (to confirm the swelling is a thyroid)
8. Palpate from behind (for contour/ nodules)
9. Palpate for cervical /axillary nodes
10. Ascultate the precordium for murmurs
11. Ascultate the thyroid for bruit
12. Thank the patient for allowing to examine .

OSCE 4 :

COUNSELLING

1. Inform this patient regarding his thyroid function tests which shows primary hypothyroidism .
2. What will you mention regarding the prognosis of the condition. (treated and untreated)
3. Explain how to take the drug and at what time you will ask the patient to take and how many times they have to take the drug per day .
4. Name some dietary substances / drugs to be avoided while taking the drug .
5. Explain the patient about symptoms of overdosing and underdosing so that he/she can come at the right time to you.



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OSCE 5 :

A 43 yr old smoker with previous history of treated Pulmonary tuberculosis comes to you with complaints of shortness of breath for 1 hour duration . On examination he has absent breath sounds on the right hemithorax with tracheal shift to the left and hyperresonant note on right hemithorax on percussion.

Based on the data given :

1. What is the likely diagnosis ?
2. Name four etiologies for the condition .
3. What is the first investigation you would like to consider ?
4. If the patient has hypotension along with the above findings what will be the first intervention you would like to do to this patient ?
5. If the patient comes to you multiple times with the same condition , name two treatment options to prevent future recurrences.

ANSWER :

1. Right sided Pneumothorax
2. COPD, TB, Smoking , Cystic fibrosis , Staphylococcal pneumonia
3. Chest X ray
4. Insert a wide bore needle at the second ICS
5. Pleurectomy , Pleurodesis .

OSCE 6 :

A 36 year old lady comes to you with Shortness of breath for 2 months, which worsened recently . She also complains of lethargy and fever episodes in the evening. On examination she was tachypneic and absent breath sounds in the left hemithorax along with tracheal shift to the right and stony dull note on percussing the left hemithorax . Based on the data :

1. What is the most likely diagnosis ?
2. What is the most common etiology in India ?
3. What is the first investigation that will confirm your diagnosis ?
4. What will be the best investigation to determine the etiology ?
5. Name one potential therapeutic option that might save the life of this patient .

ANSWER :

1. Left sided Pleural effusion
2. Tuberculosis
3. Chest X ray PA view
4. Diagnostic thoracocentesis / pleural fluid analysis
5. Intercostal chest drain.

OSCE 7 :

Examine the integrity of cerebellum in this patient.

1. Eyes (nystagmus)
2. Limbs (hypotonia)
3. Speech (dysarthria)
4. Gait (ataxia)
5. Signs
 - Dysdiadokinesia
 - Finger nose test
 - Rebound phenomenon
 - Heel shin knee test
 - Pendular knee jerk .

OSCE 8 :

You are viewing this patient who has a five day history of fever and his Blood smear shows RBC's positive for *P. falciparum*.

1. Name the type of fever *P. falciparum* causes
2. Name two complications of the disease
3. Name two drugs that will be effective in treating *P. falciparum*
4. Name the drug that should be given very carefully in patients with G6PD deficiency.
5. Name two drugs that can be used for chemoprophylaxis of malaria.

ANSWER :

1. Malignant tertian malaria
2. Black water fever , Cerebral malaria
3. Artemether , Artether , Lumefantrine+artemether , Quinine
4. Primaquine
5. Mefloquine , Doxycycline

OSCE 9 :

A 46 yr old gentleman who is a known diabetic for 14 years on insulin is admitted with complains of fever , cough with expectoration and shortness of breath of 4 days duration which has worsened so much in the last 8 hours . On examination he was tachypneic and there was crepitations and Bronchial breath sound heard In right mammary and infrascapular regions. Based on the data given

1. What is the likely clinical diagnosis ?
2. Name two investigations you would like to do it first ?
3. Name two commonly associated organisms.
4. Name two risk factors for developing the condition.
5. What is the treatment of choice if the associated organism is MRSA ?

ANSWER :

1. Community acquired pneumonia
2. Chest X ray , Sputum C/S
3. Pneumococcus, Hemophilus influenza
4. Immunodeficiency (HIV, steroids , transplant), Diabetes , Influenza(post)
5. Vancomycin / Linezolid

OSCE 10 :

Examine this patient who has probable splenomegaly.

STEPS :

1. Introduce yourself
2. Explain what you are going to do and obtain a verbal consent
3. Stand to the right of the patient and ask him/her to strip
4. Inspect for visible fullness in Left hypochondrium
5. Palpate from RIF towards the left hypochondrium . If ascites use dipping method .
6. Observe patient's face for pain when palpating the patient's abdomen. Enquire him/her periodically about the same .
7. Correct technique of palpation (Hands in line with the wrist and leaning forward)
8. Thanking the patient for cooperating .

OSCE 11 :

A 46 year old chronic alcoholic (>20 yrs) comes to you with abdominal distension and pedal edema of 3 months duration. He also complains of yellowish discoloration of urine and eyes. On examination , he was deeply icteric, pale and has bilateral pedal edema . He also had tense ascites with fluid thrill.

Based on the data provided :

1. What is the likely diagnosis ?
2. Name two signs of liver cell failure .
3. Name two Investigations which will be helpful in arriving at the diagnosis
4. How will you calculate SAAG ?
5. Name two complications of portal hypertension.

ANSWER :

1. Decompensated Chronic Liver Disease – Ethanol related
2. Spider naevi, Palmar erythema , Dupuytren's contracture
3. Liver Function Tests and Ascitic fluid analysis or USG abdomen
4. Serum albumin – Ascitic fluid albumin .
5. UGI (variceal) bleed, Hepatic encephalopathy

OSCE 12 :

Examine the motor component of facial nerve in this patient .

EXAMINE THE MOTOR COMPONENT OF FACIAL NERVE

A 56 year old gentleman comes to emergency with severe chest pain and breathlessness of 1 hour duration. The pain started when he was eating and is retrosternal and is radiating to his left shoulder, hand and jaw. On examination he looks unwell, sweating and he has a blood pressure of 78/30 mm Hg. He also had fine crepitations in both lung bases. You take an ECG which shows ST segment elevation (5mm) in precordial leads. (V1- V5) with reciprocal changes in inferior leads (II, III, aVF)

1. What is your diagnosis ?
2. What is the next investigation you would like to do to confirm the diagnosis ?
3. Name any two treatment modalities available.
4. Name two complications.
5. Name two drugs that will be beneficial in long term .

Examine this patient who has jaundice

Examine this patient for signs of liver cell failure.

Examine this patient with anaemia .

A 45 year old lady comes to OPD with 2 day history of fever, burning sensation while passing urine and lower abdominal pain which radiates to the back . She is a known diabetic for >10 years and she on Metformin and glipizide for 6 years. Her recent Fasting blood sugar is 141 mg/dl and HbA1c is 7.1%. On examination , she is febrile and has mild tenderness on palpating the hypogastrium .

1. What is the likely diagnosis ?
2. What is the commonest organism ?
3. Name two investigations useful in this patient to confirm diagnosis .
4. Name two risk factors for the disease.
5. Name two drugs useful in treating the disease.

7



1. Diagnosis?
2. What are the eyes signs of this condition?
3. What is thyroid acropathy?

8



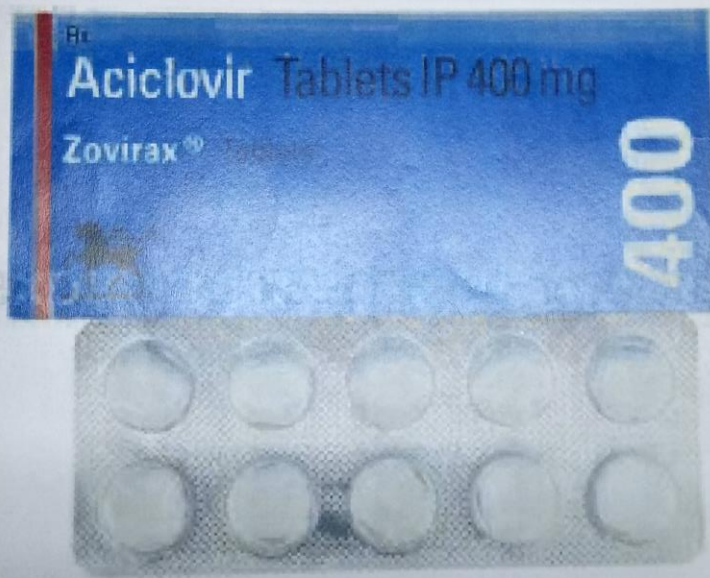
1. What is heimlich's manoeuvre
2. Explain - BVM
3. What is the ratio of chest compression to respiration in Basic life support with single rescuer?
4. How much air has to be delivered to a normal male adult during resuscitation?

12. This is the laboratory report of a 55 year old man who had poorly controlled diabetes and hypertension for the last 20 years.

- pH 7.1
- pCO₂ 20 mmHg
- pO₂ 90 mmHg
- HCO₃ 10 mEq/l
- Blood glucose 476 mg/dl
- Potassium 6.1 mEq/l
- Sodium 132 mEq/l
- Blood urea 210 mg/dl

- › a) What acid base disorder is present?
- › b) What is the underlying cause of these metabolic abnormalities?
- › c) List two definitive treatments for the underlying condition.

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1. what are the effects of viral meningitis in the CSF picture.
2. where does the herpes virus remain in the latent stage
3. Name the Smear that is useful in making a diagnosis
4. What are the potential side effects of this drug.

13

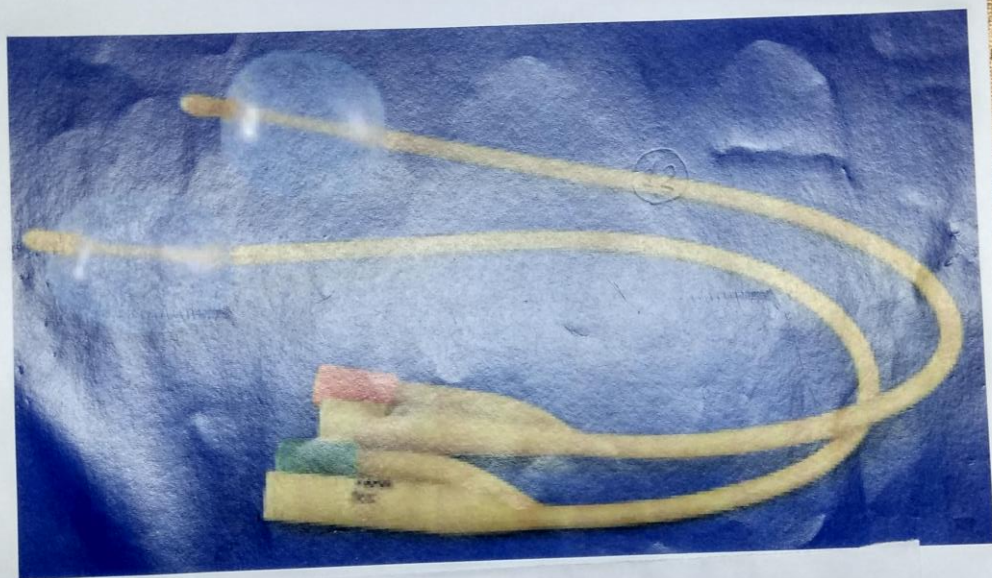
1. 45 yrs old male type II diabetes admitted with

- Blood sugar 454 mg
- Urine acetone positive
- PH 7.1
- HCO_3^- 8m Eq/lit

Question

1. What are the types of acetone bodies?
2. What is the type of breathing in this condition?
3. Which type of insulin do you prefer in this condition?
4. What are the electrolyte abnormalities of this condition.

11



5,

1, What is the diameter of 16 Fr Foley Catheter?

5mm.

How many ports does it have?

- 2

What are the complications of prolonged use?

UTS, cysts

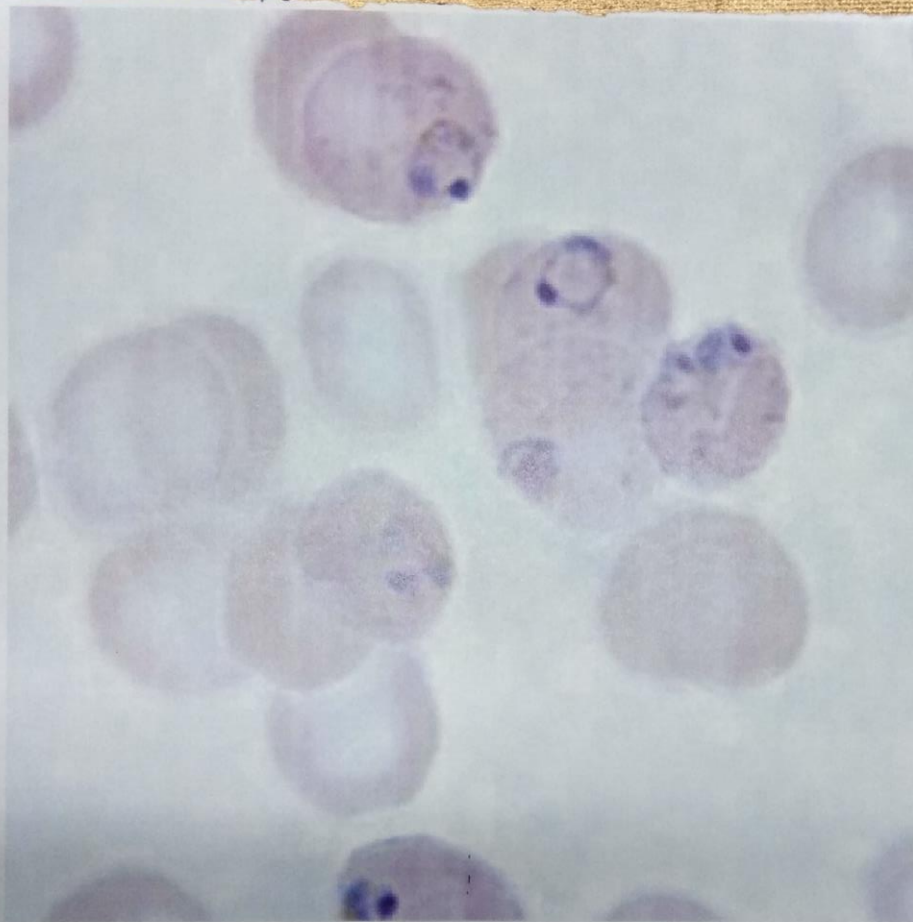
What is significant bacteruria in females

$>10^5$

5



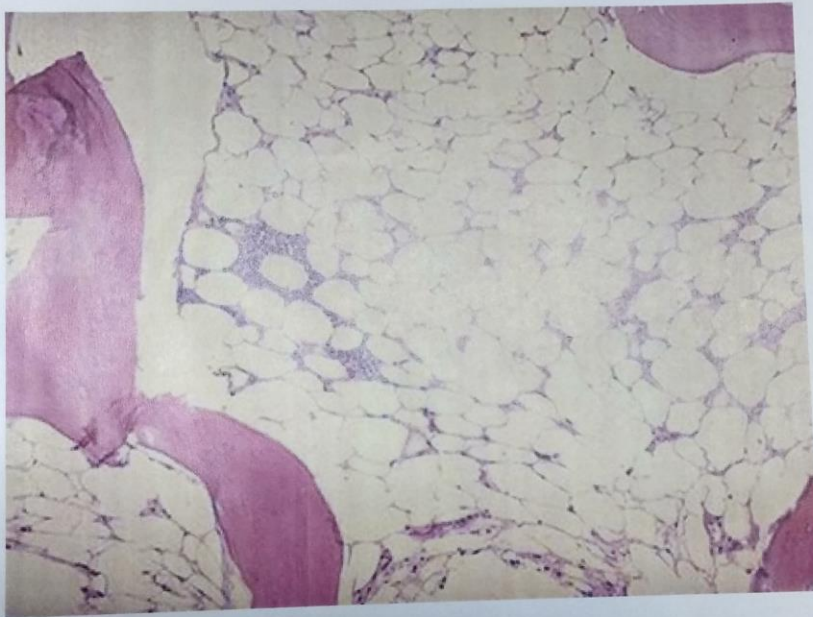
1. What is your diagnosis?
2. What are the cardiac causes of this condition?
3. Management of squatting episodes?



Peripheral Blood Smear

1. Diagnosis?
2. What is Tropical splenomegaly syndrome?
3. Which species is quarten malaria found

16



Bone marrow aspiration

1. What is your diagnosis?
2. Etiology? of this condition
3. How is the spleen in this condition?



1. what are the types of cholinergic receptors?
2. what are the signs of atropinisation?
3. In which poisoning is atropine widely used
4. What is its action on the eyes

You are viewing this patient who has a five day history of fever and his Blood smear shows RBC's positive for *P. falciparum*.

1. Name the type of fever *P. falciparum* causes
2. Name two complications of the disease
3. Name two drugs that will be effective in treating *P. falciparum*
4. Name the drug that should be given very carefully in patients with G6PD deficiency.
5. Name two drugs that can be used for chemoprophylaxis of malaria.

PLEURAL FLUID ANALYSIS

MACROSCOPIC APPEARANCE :

STRAW COLOUR

MICROSCOPIC APPEARANCE :

PLENTY OF LYMPHOCYTES

BIOCHEMICAL :

PROTEIN – 3.5 gms

GLUCOSE – 60 gms

COMMENTS

A 46 year old gentleman who is a known diabetic for 14 years on insulin is admitted with complains of fever, cough with expectoration and shortness of breath of 4 days duration which has worsened so much in the last 8 hours . On examination he was tachypnoeic and there was crepitation and Bronchial breath sound heard In right mammary and infra scapular regions. Based on the data given

1. What is the likely clinical diagnosis ?
2. Name two investigations you would like to do it first ?
3. Name two commonly associated organisms.
4. Name two risk factors for developing the condition.
5. What is the treatment of choice if the associated organism is MRSA ?

1. Community acquired pneumonia
2. Chest X ray , Sputum C/S
3. Pneumococcus, Hemophilus influenza
4. Immunodeficiency (HIV, steroids ,
transplant), Diabetes , Influenza(post)
5. Vancomycin / Linezolid

A 26 year old lady comes to you with complaints of jaundice and fatigue of 3 months duration. On examination, patient was pale and had moderate splenomegaly. Investigations revealed, Hemoglobin- 3.1 g/dl, Serum Bilirubin – 4.8 mg/dl with 98 % being Indirect bilirubin . Reticulocyte count – 11.4% (normal – 0.5-1.5) .

1. What is your likely diagnosis?
2. Name two causes
3. Name two relevant investigations
4. Mention two causes of massive splenomegaly.
5. Name atleast two things that you would like to look in the peripheral smear of this patient.

A 43 year old smoker with previous history of treated Pulmonary tuberculosis comes to you with complaints of shortness of breath for 1 hour duration . On examination he has absent breath sounds on the right hemi thorax with tracheal shift to the left and hyper resonant note on right hemi thorax on percussion. Based on the data given:

1. What is the likely diagnosis?
2. Name four etiologies for the condition.
3. What is the first investigation you would like to consider?
4. If the patient has hypotension along with the above findings what will be the first intervention you would like to do to this patient?
5. If the patient comes to you multiple times with the same condition, name two treatment options to prevent future recurrences.

A 21 year old lady is coming to you with complaints of lethargy, constipation and irregular menses . She also admits she has been intolerant to cold off late and that she gained a substantial amount of weight in recent times. On examination she was obese (BMI -31) , has a BP- 130/110 mm Hg, PR- 48/min , and a smooth butterfly shaped swelling in the anterior aspect of neck.

Based on the data :

1. What is your diagnosis?
2. What blood test you will do to diagnose the above condition?
3. Name two skin changes associated with the condition?
4. Name two complications of the disease.
5. What is the treatment of choice for the above condition and how will you give?

A 36 year old lady comes to you with Shortness of breath for 2 months, which worsened recently . She also complains of lethargy and fever episodes in the evening. On examination she was tachypneic and absent breath sounds in the left hemi thorax along with tracheal shift to the right and stony dull note on percussing the left hemi thorax . Based on the data :

- 1.What is the most likely diagnosis?
- 2.What is the most common etiology in India?
- 3.What is the first investigation that will confirm your diagnosis?
- 4.What will be the best investigation to determine the etiology ?
- 5.Name one potential therapeutic option that might save the life of this patient .

3

A 16 year old girl who is a known case of type 1 diabetes admitted to the emergency with unconsciousness and seizures since half an hour . Her class mate told that she felt dizzy while talking and she was sweating before she collapsed and threw seizures and in the process she injured her left shoulder and forehead and was biting her tongue vigorously followed by which she had passed urine and faeces. She was taking her medicines (insulin) regularly and her mother also complained that she skipped her breakfast that day morning. On examination , she was *not* seizing actively now and she was deeply unconscious with response only to painful stimuli. Her cardiovascular , respiratory and other system examinations appear normal and her vitals are stable.

1. What is the likely diagnosis ?
2. What is the next investigation you would like to do to confirm the diagnosis ?
3. What is first treatment you want to give even before the investigation results (pertaining to the case) ?
4. Name two reasons from the history to tell that it is not pseudoseizure (hysteria)
5. Name two drugs used in the status epilepticus protocol.

A 48 year old gentleman comes to OPD for routine check up. He had no complaints except for occasional headaches. He is a non-smoker and non-alcoholic. His family history is non-significant. He had no comorbidities till date. On examination he had no abnormalities except for a Blood pressure of 176/98 mm Hg. You diagnose him to be hypertensive.

1. What is the most common cause of hypertension in the world?
2. Name two causes of secondary hypertension.
3. Name two complications of hypertension if left untreated.
4. Name two drug classes useful in treating the condition.
5. Name any two relevant / must tests you would like to do in this case.

A 56 year old gentleman comes to emergency with severe chest pain and breathlessness of 1 hour duration. The pain started when he was eating and is retrosternal and is radiating to his left shoulder , hand and jaw. On examination he looks unwell , sweating and he has a blood pressure of 78/30 mm Hg. He also had fine crepitations in both lung bases. You take an ECG which shows ST segment elevation (5mm) in precordial leads. (V1- V5) with reciprocal changes in inferior leads (II, III. aVF)

1. What is your diagnosis ?
2. What is the next investigation you would like to do to confirm the diagnosis ?
3. Name any two treatment modalities available.
4. Name two complications.
5. Name two drugs that will be beneficial in long term .

A 30 year old gentleman comes to OPD with complaints of retrosternal discomfort of 4 days duration. He admits that he had retrosternal pain which was non radiating and associated with burning sensation especially in the night immediately after food. He had a fracture 1 week back and he was taking some pain killers prescribed by an orthopaedician . On examination he is well , vitals are stable and he had mild tenderness in the epigastric region.

1. What is the likely diagnosis and the cause in this patient ?
2. How will you treat and name any two drugs that can be used in treating this patient .
3. Name two potential complications of the condition.
4. Name two drugs that are active against H.pylori .
5. Name one investigation useful in confirming the diagnosis .

A 34 year old lady comes to you in emergency with complaints of shortness of breath and palpitations of 5 days duration. She is a known case of Mitral stenosis for 15 years and is on penicillin prophylaxis and she was previously well except for the last 5 days. On examination she is dyspneic, the pulse is feeble and irregular, BP- 100/60 mm Hg and auscultation revealed a mid diastolic murmur associated with a palpable diastolic thrill. She also has bi-basal crepitations in both lung bases.

1. What do you think is the likely cause of her recent decompensation ?
2. What are the characteristics of pulse and JVP you would expect in this patient ?
3. Name two important complications associated with the condition.
4. Name any two treatment modalities/drugs used to treat the condition.
5. Name two causes other than Valvular heart diseases .

A 48 year old gentleman comes to OPD for routine check up. He had no complaints except for occasional headaches. He is a non-smoker and non-alcoholic. His family history is non-significant. He had no comorbidities till date. On examination he had no abnormalities except for a Blood pressure of 176/98 mm Hg. You diagnose him to be hypertensive.

1. What is the most common cause of hypertension in the world ?
2. Name two causes of secondary hypertension.
3. Name two complications of hypertension if left untreated.
4. Name two drug classes useful in treating the condition.
5. Name any two relevant / must tests you would like to do in this case.

A 29 year old female presented with weakness, palpitation and shortness of breath on exertion..laboratory investigation reveals

- Hb 8.5 ~~mg~~/dl
- RBC,WBC and platelet counts are within normal limits
- Peripheral blood smear shows hypochromia
- MCV 65 fl

1. What is the haematological diagnosis?
2. Mention 2 causes
3. What is the commonest cause of iron deficiency anaemia?
4. What further investigations would you do?
5. How will u treat this patient

A 12-year old boy presents with scanty micturition and puffy face..Urine examination as follows

- Colour:Smoky
- Albumin +
- Pus cells 0-2 /HPF
- RBC 20-30/HPF
- RBC Cast :present

1. What is the probable diagnosis?
2. Mention three clinical signs you should look for..
3. Mention two further investigations..
4. Mention two complications
5. How would you like to treat the patient?

A 19 year old man presented with high grade fever for 2 days and disorientation for 4h. Neck rigidity and Kernig sign are present..CSF examination shows

- Colour: Turbid
- Pressure: Increased
- Biochemistry: Protein 190 mg/dl, sugar 16 mg/dl, chloride 670 mg/dl
- Cytology: total cells 6200/cmm, polymorphs 96%, lymphocytes 4%

1. What is the likely diagnosis?
2. What is the commonest organism?
3. What are the complications?
4. If the patient develops shock, what is the cause?
5. What would you look for before doing lumbar puncture?

A 46 year old chronic alcoholic (>20 yrs) comes to you with abdominal distension and pedal edema of 3 months duration. He also complains of yellowish discoloration of urine and eyes. On examination, he was deeply icteric, pale and has bilateral pedal edema. He also had tense ascites with fluid thrill. Based on the data provided:

1. What is the likely diagnosis?
2. Name two signs of liver cell failure.
3. Name two Investigations which will be helpful in arriving at the diagnosis
4. How will you calculate SAAG?
5. Name two complications of portal hypertension.

A 35 year old female presented with generalised weakness and weight gain.. Her thyroid function tests reveal

- Serum FT_3 : 2 pmol/L (normal 1.3 -3.5 pmol/L)
- Serum FT_4 : 7 pmol/L (normal 70-160 pmol/L)
- TSH: 35 mIU/L (normal 0.5-5.1 mIU/L)

1. What is the most likely clinical diagnosis?
2. Mention 2 clinical signs that you could find in this patient
3. What are the ECG changes you can expect in this patient?
4. Suggest further investigations for this patient.
5. What is the treatment?

1. Hypothyroidism

1. Free T3, T4 and TSH
2. Alopecia , Dry skin , Periorbital puffiness
3. Myxedema coma, Brady arrhythmia , Pericardial Effusion/ tamponade , Hypertension
4. Levothyroxine – once a day in the morning in empty stomach

A 34 year old lady comes to you in emergency with complaints of shortness of breath and palpitations of 5 days duration. She is a known case of Mitral stenosis for 15 years and is on penicillin prophylaxis and she was previously well except for the last 5 days. On examination she is dyspneic, the pulse is feeble and irregular, BP- 100/60 mm Hg and auscultation revealed a mid diastolic murmur associated with a palpable diastolic thrill. She also has bi-basal crepitations in both lung bases.

1. What do you think is the likely cause of her recent decompensation ?
2. What are the characteristics of pulse and JVP you would expect in this patient ?
3. Name two important complications associated with the condition.
4. Name any two treatment modalities/drugs used to treat the condition.
5. Name two causes other than Valvular heart diseases .

A 43 year old gentleman presents to you with fever, shortness of breath, cough with expectoration, right sided pricking chest pain which increases with inspiration for 2 days , all of which has worsened for the past 2 hours . He is a known case of type 2 diabetes of 4 years duration and his records suggest that he is not compliant with his diabetic medications with recent fasting blood sugar of 241 mg/dl. On examination he is febrile , and had increased vocal resonance, coarse crepitations with bronchial breath sounds in right infra axillary and infra scapular areas.

1. What is your diagnosis ?
2. Name two causative organisms.
3. What is the next best investigation.
4. Name one complication which you should look for pertaining to this case.
5. If the patient is HIV positive with a CD4 count of $96/\mu\text{l}$, what other organism you should cover for while starting treatment.

INSTRUMENTS

IV CANNULA





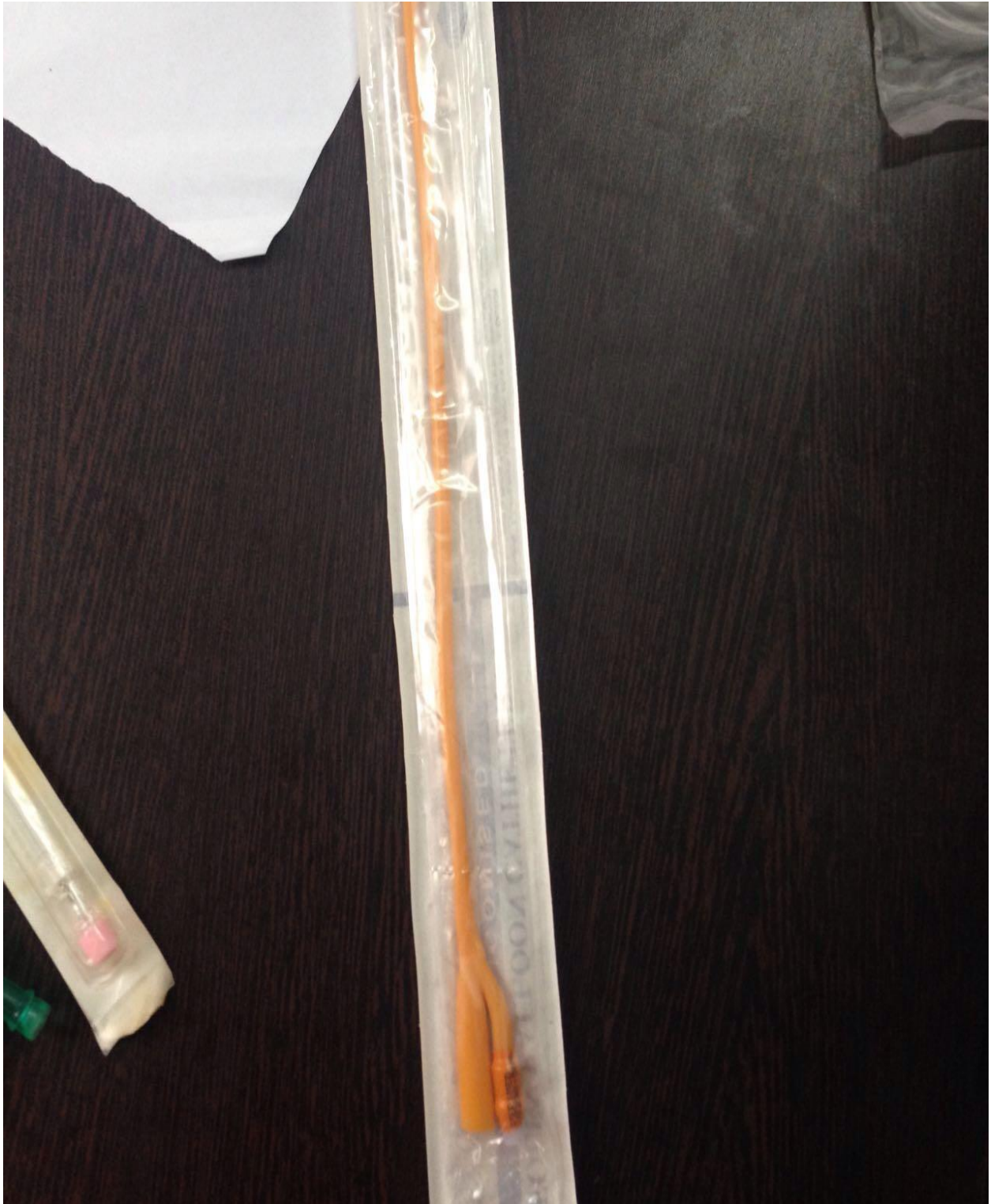




BONE MARROW ASPIRATION NEEDLE



FOLEY'S CATHETER





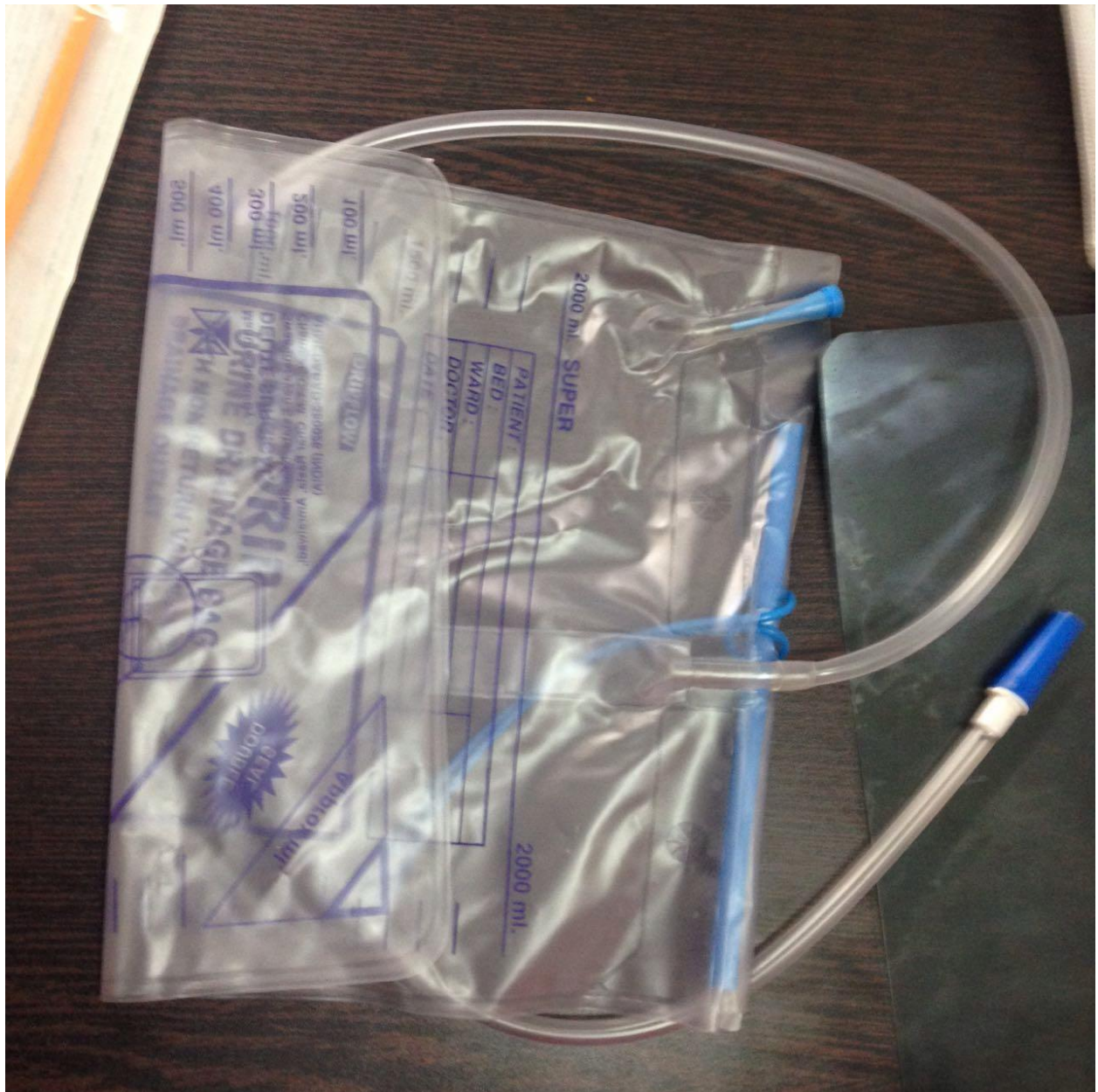
TISSUE BIOPSY NEEDLE



AMBU-BAG



URINARY BAG



RYLE'S NASOGASTRIC TUBE



ENDOTRACHEAL TUBE



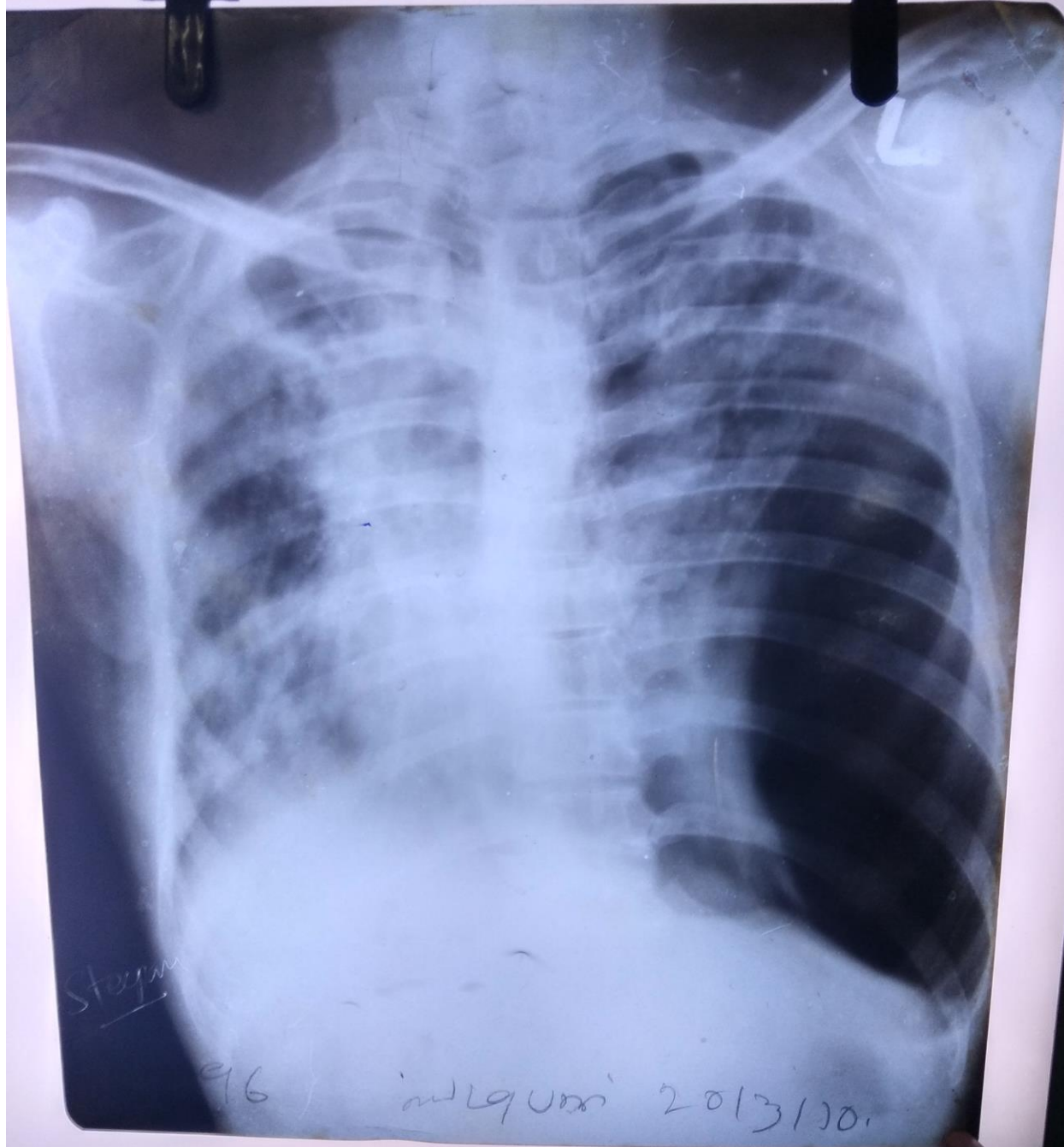
INSULIN INJECTION

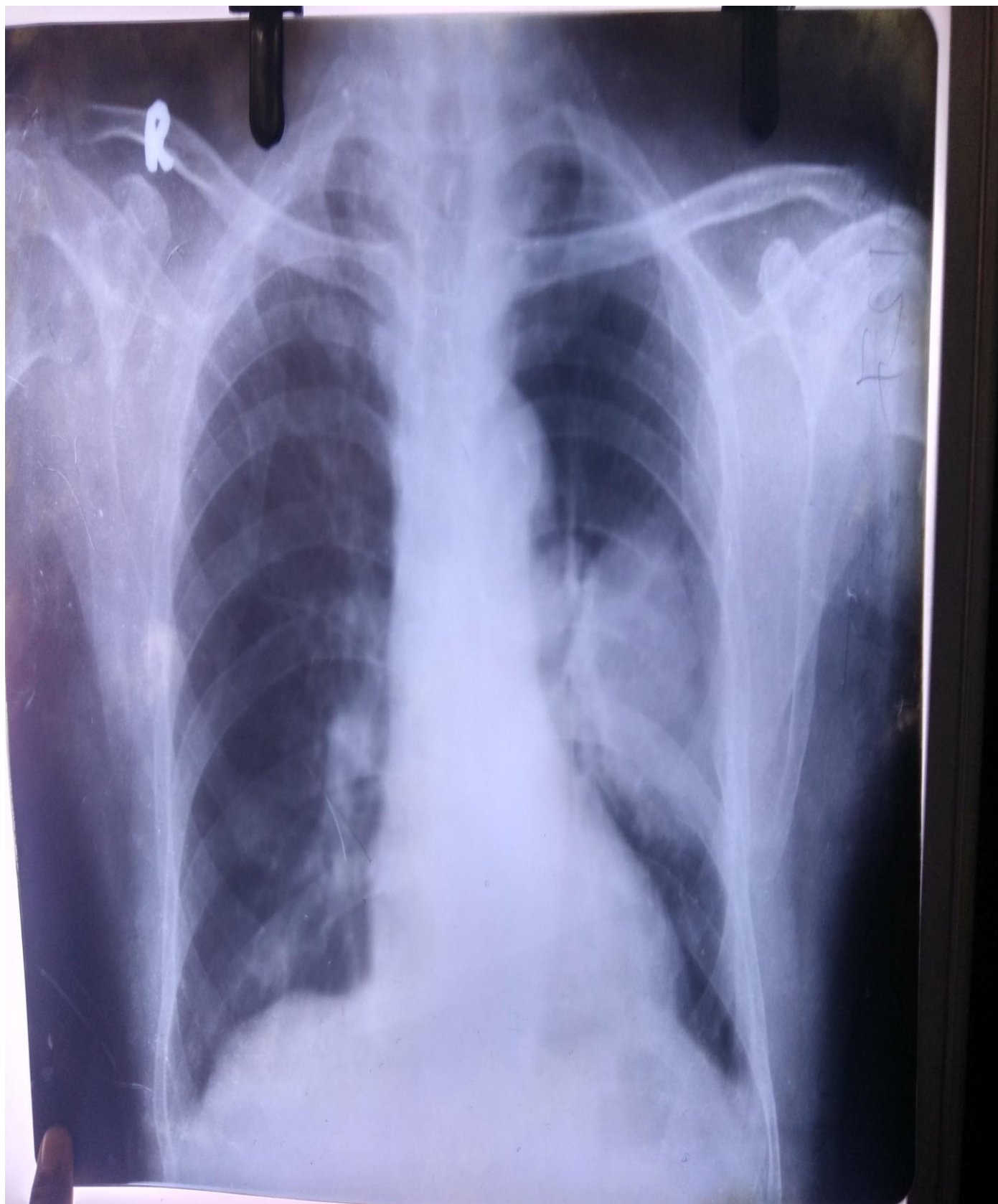


LUMBAR PUNCTURE NEEDLE



X-RAYS

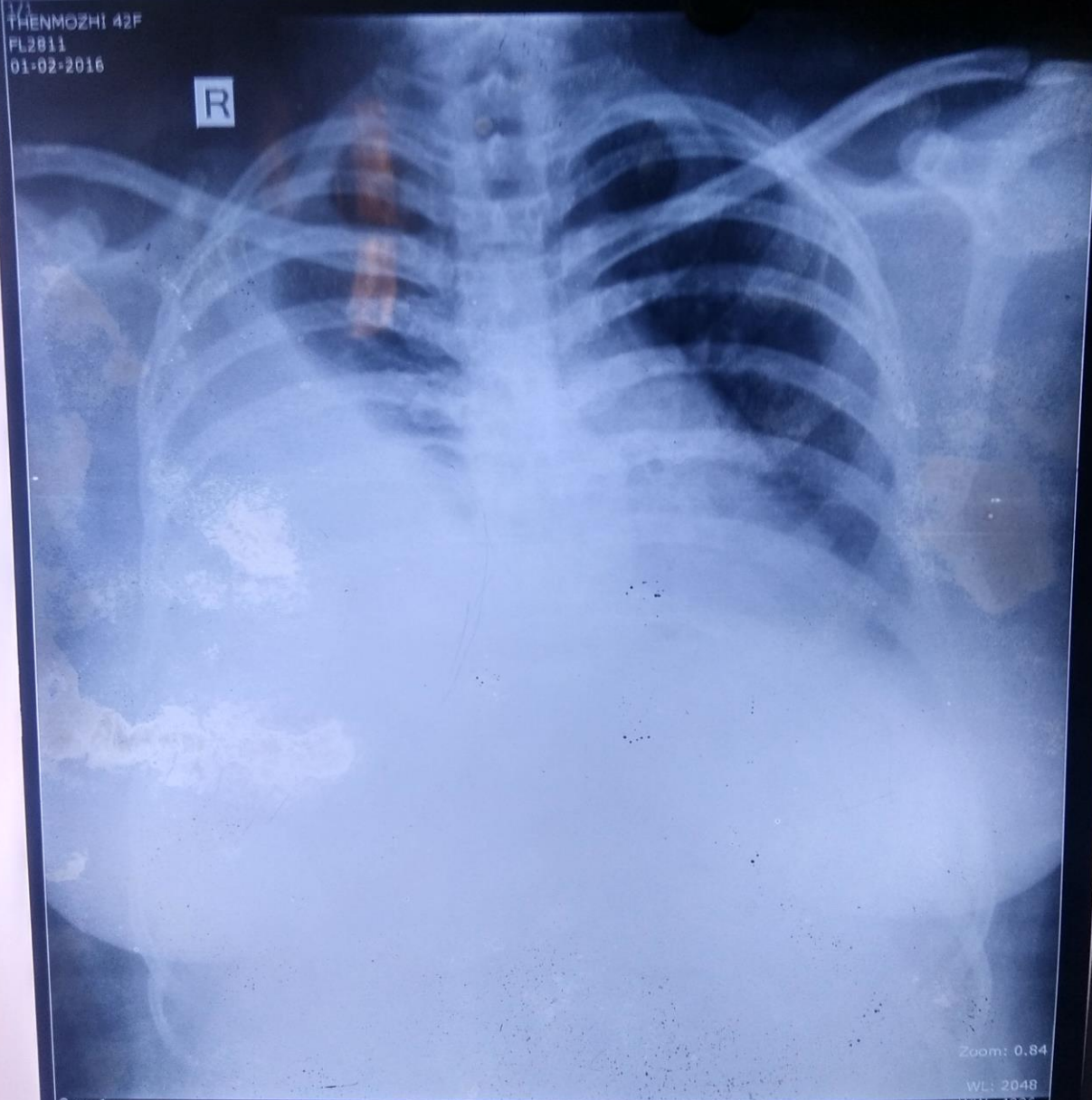




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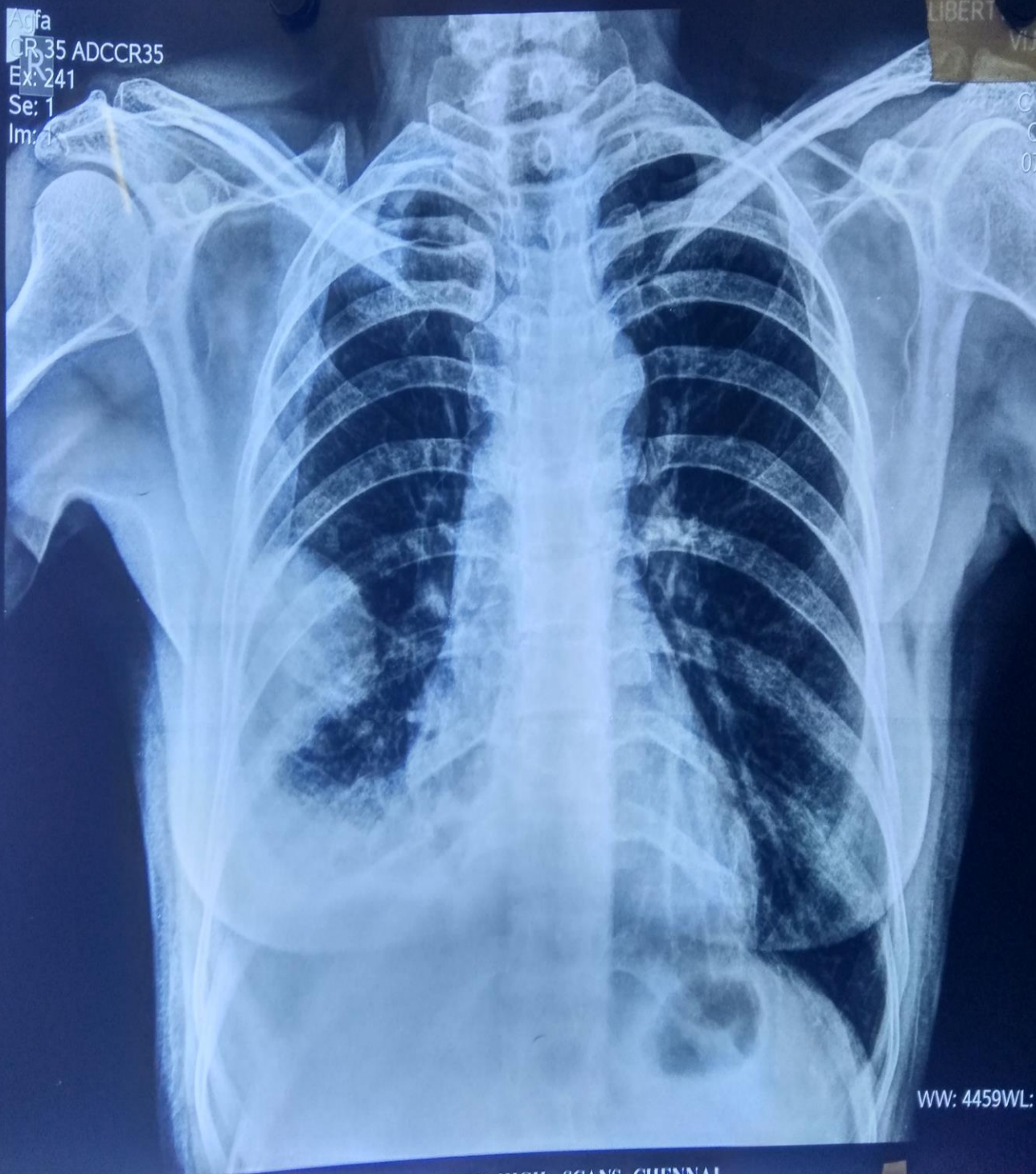


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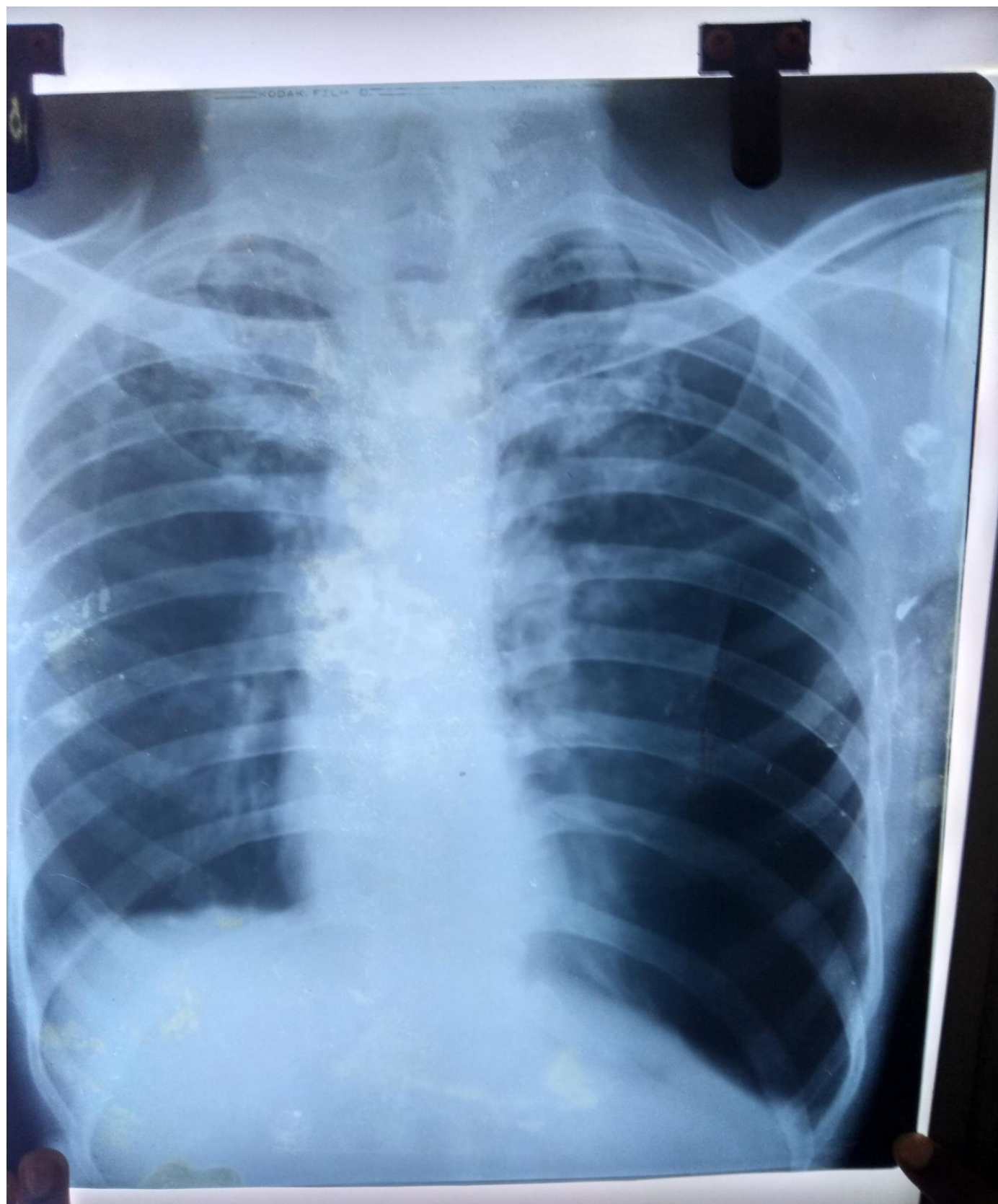
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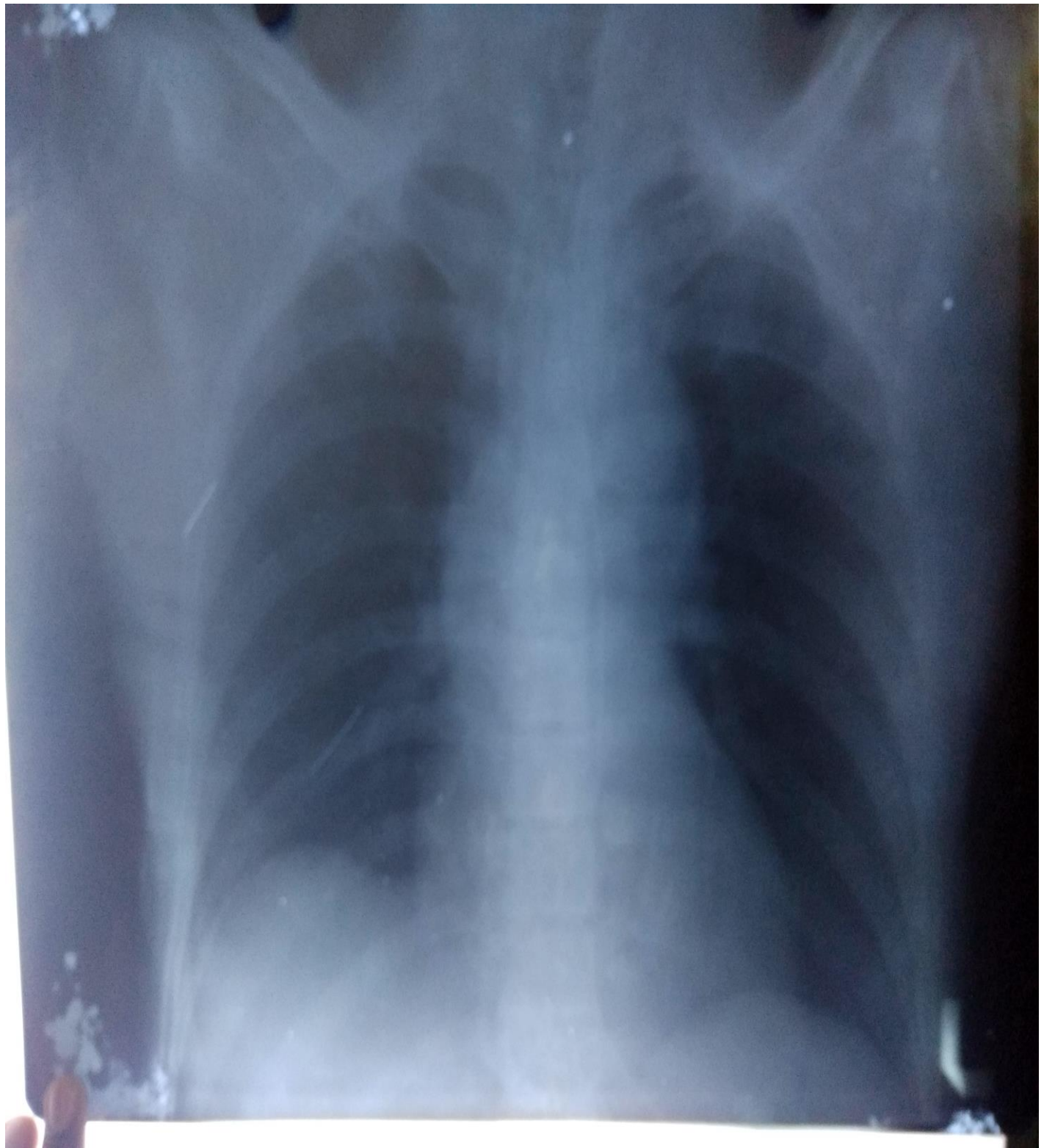


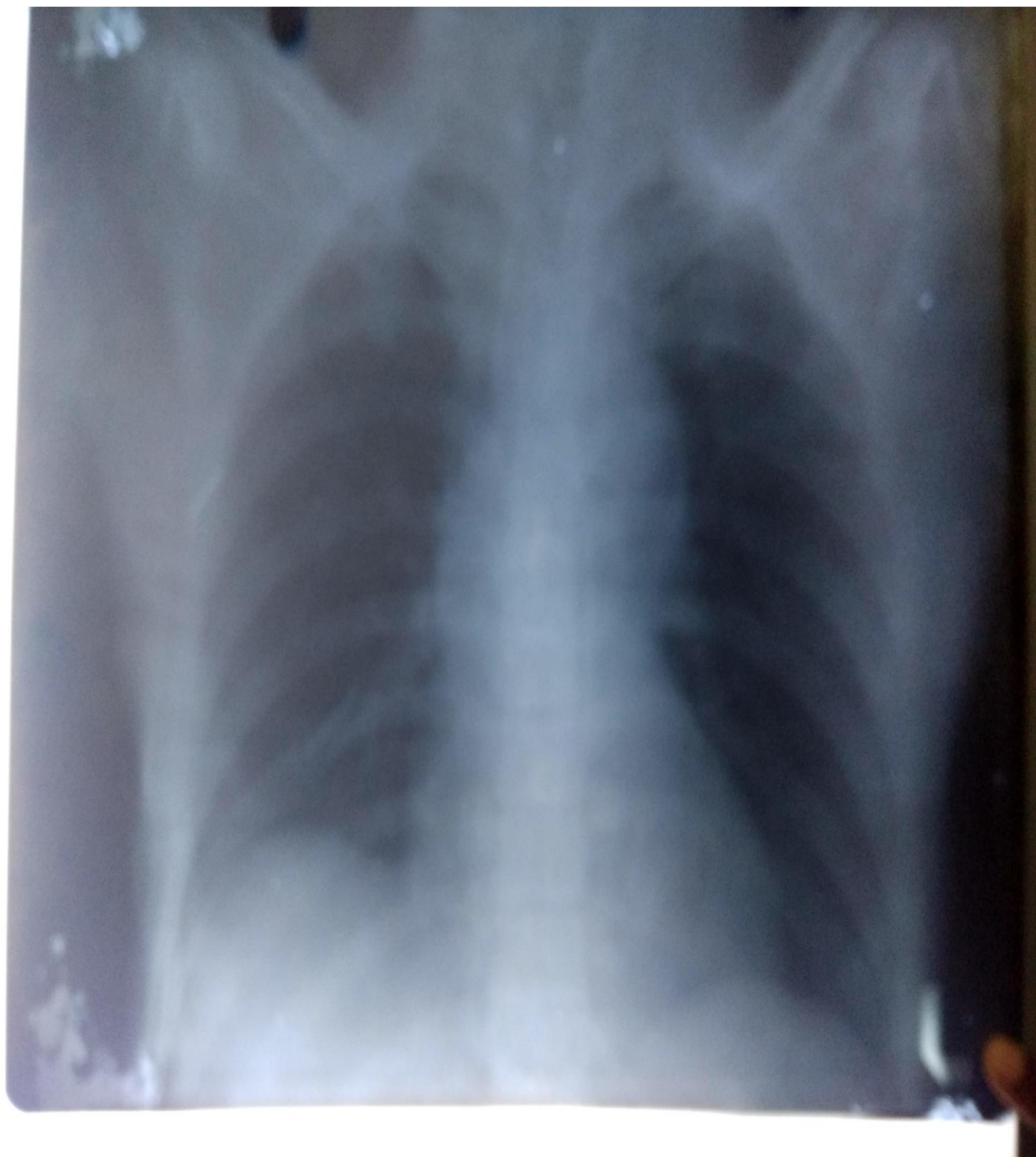
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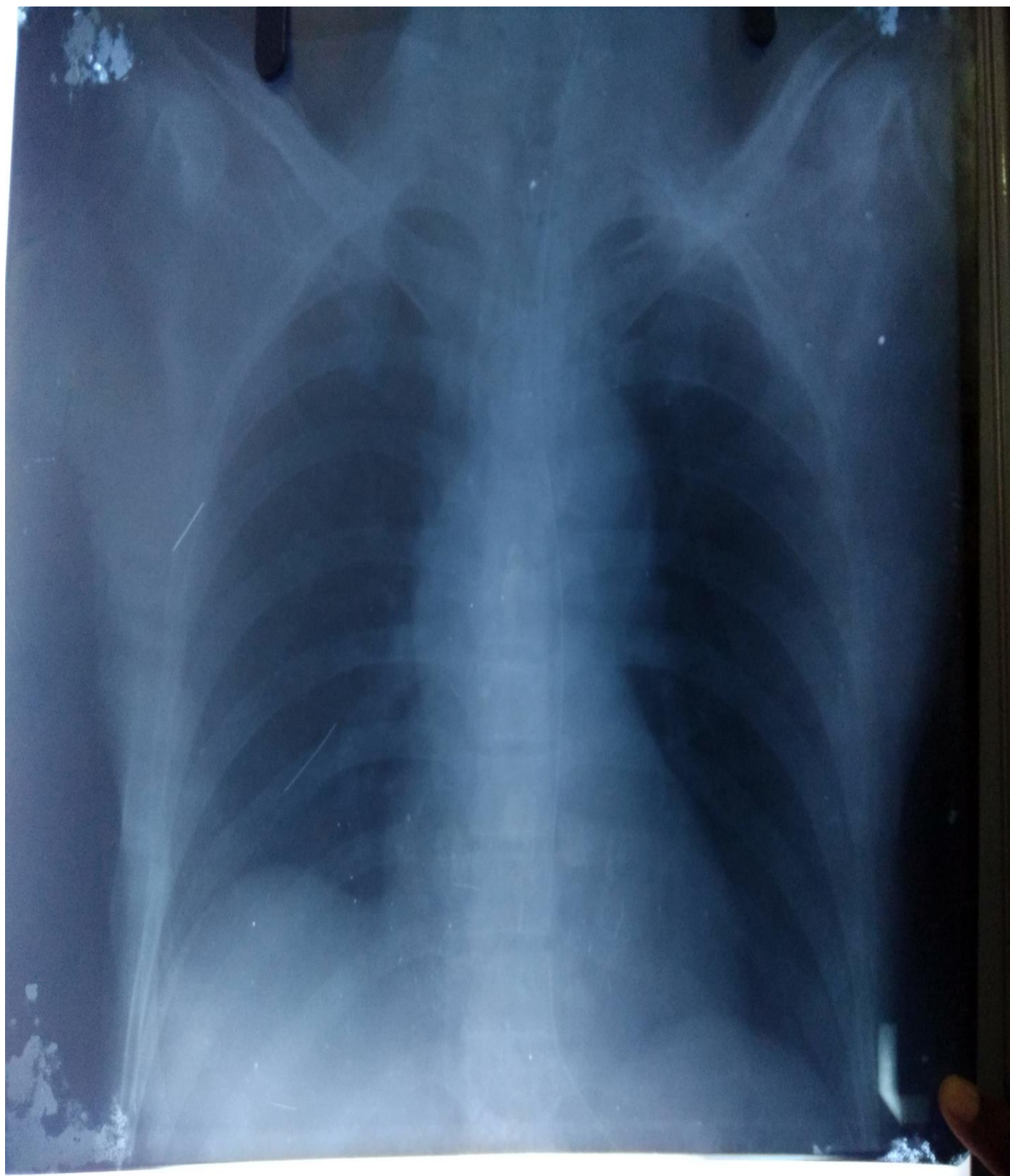
SCANS CHENNAI











VENKATA PRASAD, 34/M

15-Apr-2012

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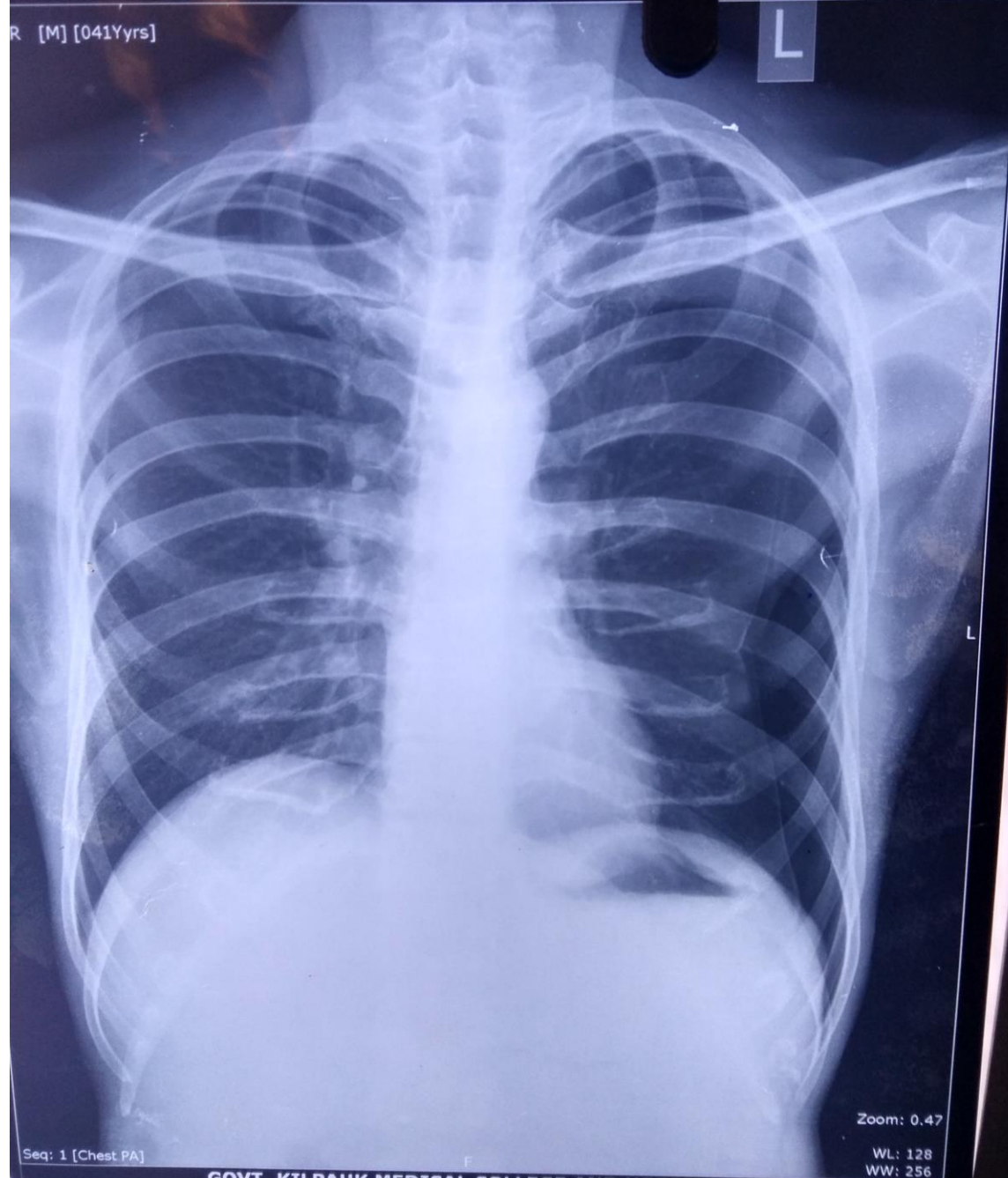
CHEST ERECT PA VIEW



DHANASEKAR / Id : AN2347

R [M] [041Yyrs]

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Zoom: 0.47

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